CPE 486/586: Deep Learning for Engineering Applications

05 Deep Learning for Sequence Data

Spring 2026

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Outline

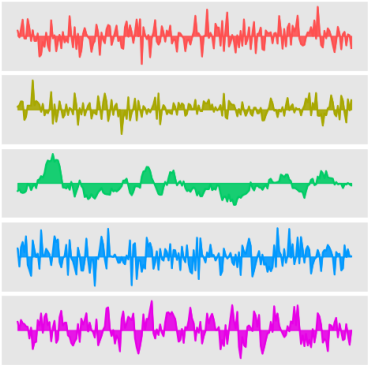
Motivation

Until Now ...

1. Data modality: Non-sequential (tabular with no order for samples or features)
2. Process modality: iterative refinement of solution, but not necessarily sequential (non-generative processes – hence no notion of sequential)

Sequential Data

Image/Video



Timeseries

Efficient modelling of feature interactions underpins

Texts/Natural Language/Computer Code

Can we use what we learnt so far on Sequential Data?

Short answer: **May be!** But not quite!

So where is the problem?

Underlying Assumptions

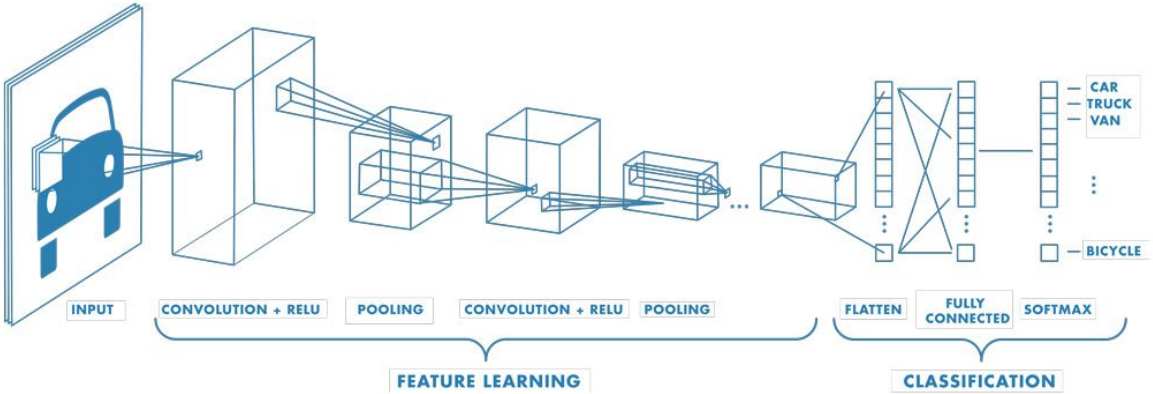
1. Core assumptions have been that data samples are iid (independent and identically distributed)
2. There are also no temporal or spatial dependencies (order of features in the tabular data didn't matter until now).
3. Generally feature length (or number of features are fixed).
4. Non-stationarity assumption: data distribution didn't change.

Learning on Sequential Data

1. Images: Convolutional Neural Network (CNN)
2. Texts/Natural language: Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM) Network, Gated Recurrent Unit (GRU) Network, Transformer
3. Videos: CNN + LSTM/GRU/Transformer

Convolutional Neural Network

Convolutional Neural Network

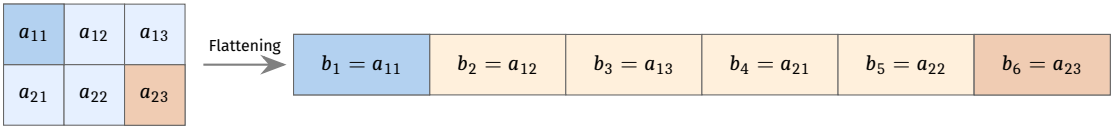


Source:

https://www.mathworks.com/discovery/convolutional-neural-network.html

Motivation for CNN

- **(Fully-connected Neural Network, FNN Limitation:** Spatial data must be converted to a **1-D array** (Flattening).



- × **Spatial Loss:** We lose the pixel-to-pixel relationship and local geometry.
- × **RGB Complexity:** In addition, working with color images would mean we have at least three channels, R, G, B.
- × **Parameter Explosion:**
  - $200 \times 200 \times 3 = \mathbf{120,000}$ features per image.
  - Just 100 images = **12,000,000** features to process!

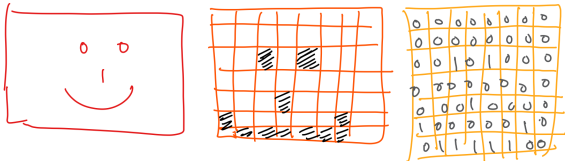
Limitations of FNN for Images

In that sense, FNN suffers from two conditions:

- Unable to preserve spatial information
- Curse of dimensionality

To preserve the pixel-to-pixel spatial relationship, we include two additional operations in the neural network:

1. Convolution
2. Pooling



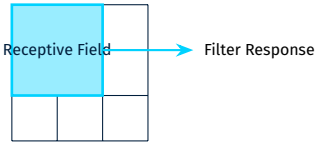
What is a CNN?

Beyond Flattening

A **Convolutional Neural Network (CNN)** is designed for grid-structured data. Unlike FNNs, it treats images as spatial volumes, preserving local relationships.

The Architectural Edge:

- **Local Receptive Fields:** Neurons only “see” a small local region, mimicking the human visual cortex.
- **Spatial Hierarchy:** Layers learn to recognize edges, then shapes, then objects.



Weight Sharing & Pooling

Weight Sharing

Filters share weights across the entire image.

- **Efficiency:** Drastically reduces parameters.
- **Feature Detection:** A vertical edge detector works everywhere!

Pooling

Subsampling to reduce spatial dimensions.

- **Invariance:** Stable against small translations/shifts.
- **Speed:** Lowers computational load & memory.

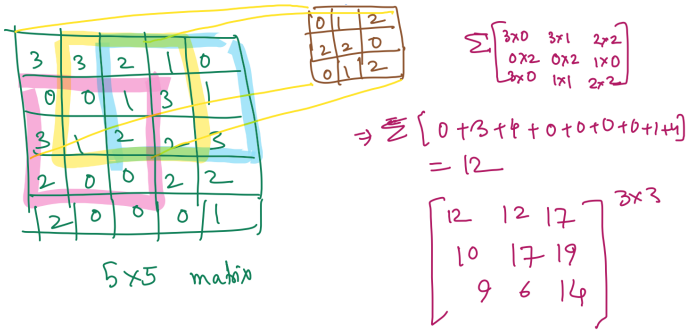
FNN: 224x224 image $\approx$ 150k+ weights per neuron.
CNN: 3x3 filter = **9 weights** per feature map.

Building blocks: Convolution

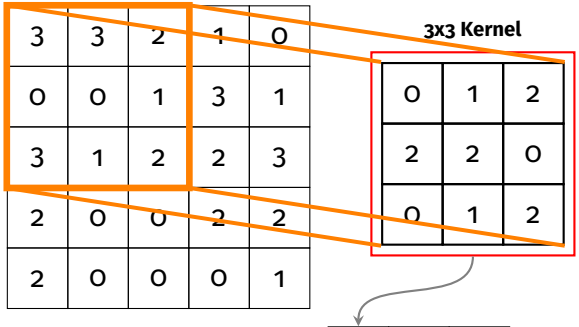
Convolution is an operation to reduce the number of pixels through some operation that, in turn, reduces the number of weights to learn.

In convolution, the reduction in the number of features happens by sliding a kernel across an image, performing the dot product.

Kernel: It is an $n \times n$ matrix of some numbers whose dimension is usually smaller than the image size.



Convolution Using Kernel



$$\sum \begin{bmatrix} 3\times0 & 3\times1 & 2\times2 \\ 0\times2 & 0\times2 & 1\times0 \\ 3\times0 & 1\times1 & 2\times2 \end{bmatrix}$$

$$= 0+3+4+0+0+0+0+1+4$$

$$= \mathbf{12}$$

Kernels

A particular kernel usually extracts certain features from an image, or basically, it performs some sort of image processing.

Examples of Kernels

$$\text{Edges: } \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\text{Sharpen: } \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\text{Box Blur: } \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Edge-detection Example

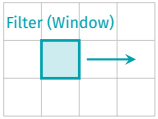
Original Image

Edge Detection Filtered Image

Filter

The window we use in kernel is called a **filter**. The window of a certain size is slid across the image. We can apply pooling to perform operations such as **minimum, median, maximum**.

- **Images** can be represented as **matrices**, where each element corresponds to a pixel.
- A **filter** is just a small matrix that is **convolved** with same-sized sections of the image matrix.



Convolutional Filters

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \cdots & \cdots \\ \vdots & \ddots & \\ \vdots & & \ddots \end{bmatrix}$$

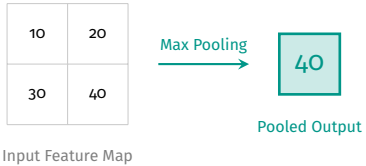
$$\begin{aligned} &(0 \times 0) + (0 \times 1) + (0 \times 0) + \\ &(0 \times 1) + (1 \times -4) + (2 \times 1) + \\ &(0 \times 0) + (2 \times 1) + (4 \times 0) = \mathbf{0} \end{aligned}$$

Convolutional Filters

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{0} & -1 & -1 & 0 \\ -2 & -5 & -5 & -2 \\ 2 & -2 & -1 & 3 \\ -1 & 0 & -5 & 0 \end{bmatrix}$$

Pooling

Pooling does a mathematical operation on the output of the convolution image. But you can also apply pooling directly to the image.



Building Blocks: Pooling

Pooling is a mathematical operation on the output of the convolution image. But you can also apply pooling directly to the image.

Pooling: Downsampling

Combine multiple adjacent nodes into a single node: max-pooling

$$\begin{bmatrix} 0 & -1 & -1 & 0 \\ -2 & -5 & -5 & -2 \\ 2 & -2 & -1 & 3 \\ -1 & 0 & -5 & 0 \end{bmatrix} \rightarrow \textbf{max} \rightarrow \begin{bmatrix} 0 & \\ & \end{bmatrix} \tag{1}$$

$$\begin{bmatrix} 0 & 1 & -1 & -0 \\ -2 & -5 & -5 & -2 \\ 2 & -2 & -1 & 3 \\ -1 & 0 & -5 & 0 \end{bmatrix} \rightarrow \textbf{max} \rightarrow \begin{bmatrix} 0 & 0 \\ & \end{bmatrix} \tag{2}$$

- Reduces the dimensionality of the input to subsequent layers and thus, the number of weights to be learned
- Further, it protects the network from (slightly) noisy inputs

Downsampling with Stride

- Only apply the convolution to some subset of the image
- e.g., every other column and row = a **stride of 2**

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2 & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

Downsampling with Stride

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2 & -2 & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

Downsampling with Stride

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 1 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

Downsampling with Stride

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 2 & 4 & 4 & 2 & 0 \\ 0 & 1 & 3 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 1 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

Operation after Pooling

- After applying convolution and pooling, we apply a non-linear activation, **ReLU**.
- Then after that, you have more Convolution, more Pooling layers.
- Finally, after several sequences of **Conv + Pool + ReLU**, we **flatten** the output and feed it to the fully connected layer (ANN).

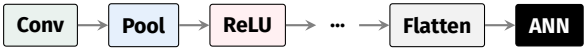
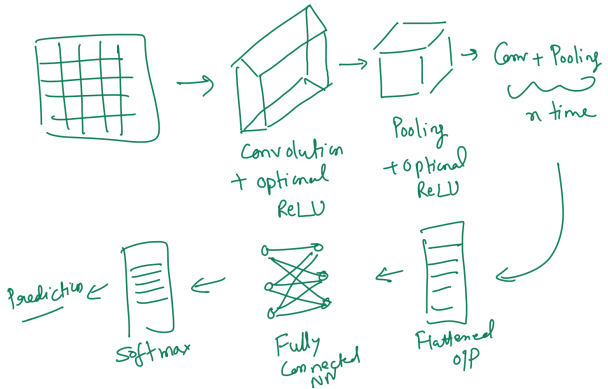
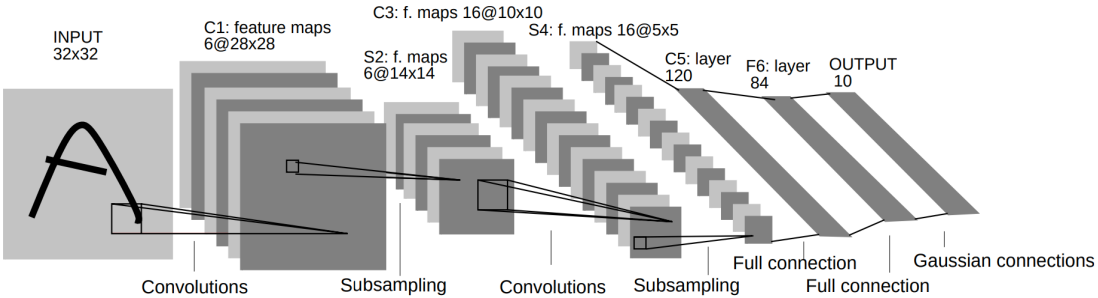


Image Classification

Then, we apply a softmax activation if our goal is to do supervised image classification.



Lenet

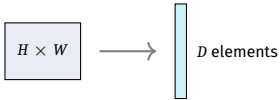


Source: http://vision.stanford.edu/cs598_spring07/papers/Lecun98.pdf

- One of the earliest, most famous deep learning models – achieved remarkable performance at handwritten digit recognition ($<$ 1% test error rate on MNIST dataset)
- Used sigmoid (or logistic) activation functions between layers and mean-pooling, both of which are pretty uncommon in modern architectures

A note on multi-dimensionality and vectorization

- **High-Dimensionality:** Images are high-dimensional datasets. This is true even more for **colored images**.
- Let $X \in \mathbb{R}^{H \times W}$ be a matrix (for a grayscale image of height H and width W).
- **Flattening:** After flattening, an image is represented as a column vector $X \in \mathbb{R}^{D}$:
  - **Grayscale:** $D = H \times W$
  - **Color (3 channels):** $D = H \times W \times 3$



Tensor

We need a concept of higher-order matrices called a **Tensor**.

- $X \in \mathbb{R}^{H \times W \times D}$ is an **order-3 tensor**. Consider it as D channels of matrices. (Note: $D = 1$ reduces it to a standard matrix).

Tensor Orders

- **Order-0:** A Scalar value.
- **Order-1:** A Vector.
- **Order-2:** A Matrix. *(Corrected from your text)*
- **Order-3:** A Cube (e.g., RGB Image).

We can also have **4 dimensions** if we consider opacity (e.g., RGBA in PNG).

Tensor to Vector

Given a tensor, we can arrange all numbers into a long vector following a **prespecified order**.

Example (Matrix Vectorization - Column First Order):

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{\text{MATLAB } A(:)} \begin{bmatrix} 1 \\ 3 \\ 2 \\ 4 \end{bmatrix}$$

Recursive Vectorization

To vectorize an **order-3 tensor**, we vectorize the first channel (matrix), then the second, and so on. This recursive process applies to tensors of any order > 3.

Tensor in CNN

In CNNs, the convolution kernel will form an **order-4 tensor**.

CNN Input Dimensions

1. **Depth:** Number of channels in a colored image.
2. **Filters:** Multiple filters learning specific features.
3. **Spatial Dimensions:** $H \times W$ of the image.

Tensor Orders (Examples):

- **Order 3:** A batch of grayscale images.
- **Order 4:** A batch of colored images or a single colored movie.
- **Order 5:** A batch of movies.

Tensors

Example:
$3 \times 4 \times 6$ tensor

4	1	2	16	3	6
1	5	2	6	14	15	8
5	26	4	6	8	9	5	3
0	0	16	5	2	8	2	1
	7	5	2	14	11	7	8
		15	2	5	0	9	8

Source: https://www.cs.cmu.edu/~mgormley/courses/10601//slides/lecture17-cnn.pdf

Further Reading

https://www.cs.toronto.edu/~hinton/absps/NatureDeepReview.pdf

http://vision.stanford.edu/cs598_spring07/papers/Lecun98.pdf

Modern Convolutional Neural Network

Modern Days of CNN Had Arduous Path

1. CNN didn't find its popularity easily.
2. There was a **lack of good datasets**.
3. **Kernel and ensemble methods** were still better until 2012.
4. Computer vision pipelines used methods such as *SIFT, SURF, bag of visual words*.
5. Researchers were using **hand-crafted features**.

The Paradigm Shift

However, **LeCun, Hinton, Ng**, and others thought features can be learnt too. **AlexNet** (after Alex Krizhevsky) proved that.

Finally Some Interesting and Useful Datasets

Large-Scale Benchmarks

1. **ImageNet (2009):** https://huggingface.co/datasets/ILSVRC/imagenet-1k
2. **CIFAR-10 & CIFAR-100:** https://www.cs.toronto.edu/~kriz/cifar.html
3. **LAION Dataset:** https://laion.ai/projects/

In addition, Yan LeCun used **MNIST**:

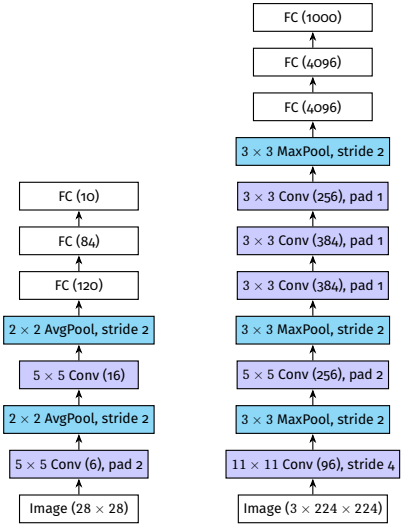
https://github.com/rahulbhadani/CPE490_590_Sp2025/tree/master/Data/MNIST

We also have **FashionMNIST**:

https://github.com/zalandoresearch/fashion-mnist

Note: These two are smaller size datasets.

AlexNet



From LeNet (left), AlexNet (right).

AlexNet Architecture

1. Convolution window shape: 11×11, then 5×5, and then 3×3.
2. Network adds maxpooling of 3×3 with a stride of 2.
3. At the end, two huge fully connected layers with 4096 outputs.
4. ReLU activation function.

Block Thinking

Neurons → Layers → **Blocks**

Basic building blocks of CNN are sequence of **(i)** Convolutional layer with padding **(ii)** a nonlinearity such as a ReLU **(iii)** a pooling layer such as max-pooling to reduce the resolution.

But pooling layer reduces dimension by $\log_2 d$ for d dimension input, and it can happen fast exponentially.

1. **Simonyan and Zisserman (2014)** used multiple convolutional layer in between downsampling via max-pooling in the form of block.
2. **Two** 3×3 **convolution** have same effect as one 5×5 convolution layer: **deep and narrow networks significantly outperform their shallow counterparts.**

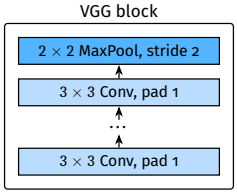
Three important papers in that area:

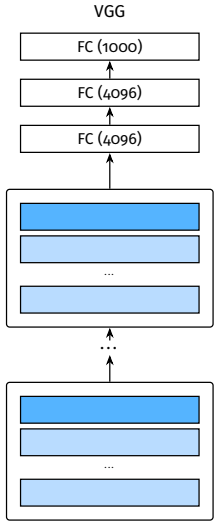
1. Simonyan, K. & Zisserman, A. "Very deep convolutional networks for large-scale image recognition." (2014).
2. Lavin, A. & Gray, S. "Fast algorithms for convolutional neural networks." (2016).
3. Liu, Z. et al. "A convnet for the 2020s." (2022).

VGG Block

VGG Block (**Simonyan and Zisserman (2014)**), consists of a sequence of convolutions with 3×3 kernels with padding of 1 (keeping height and width) followed by a 2×2 max-pooling layer with stride of 2 (halving height and width after each block).

- VGG Network can be partitioned into two parts: the first consisting mostly of convolutional and pooling layers and the second consisting of fully connected layers that are identical to those in AlexNet.
- But the convolutional layers are grouped in nonlinear transformations that leave the dimensonality unchanged, followed by a resolution-reduction step.





Network in Network (NiN)

Two main problems with AlexNet and VGGNet:

1. FCN at the end of the network has tremendous number of parameters.
2. Cannot add FCN earlier in the network otherwise it can destroy spatial information.

NiN:

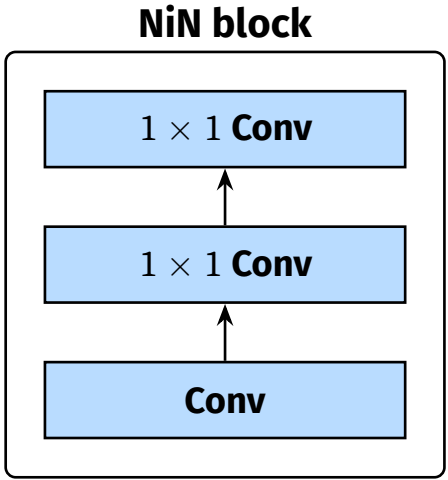
1. use 1×1 convolutions to add local nonlinearities across the channel activations
2. use global average pooling to integrate across all locations in the last representation layer

NiN Block

Input/Output of Convolutional layer is 4-d Tensor with no. of samples $\times$ channel $\times$ height $\times$ width.

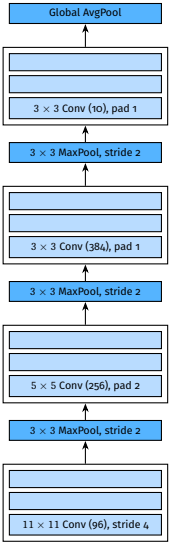
Input/output of FC layer are 2-D Tensor: no. of samples $\times$ features.

The idea behind NiN is to apply a fully connected layer at each pixel location (for each height and width). The resulting 1×1 convolution can be thought of as a fully connected layer acting independently on each pixel location.



NiN Model

- **Initial Convolution Sizes:** NiN uses the same initial convolution sizes as AlexNet:
  - Kernel sizes are 11×11, 5×5 and 3×3 respectively.
  - The numbers of output channels match those of AlexNet.
- **Downsampling:** Each NiN block is followed by a max-pooling layer with a stride of 2 and a window shape of 3×3.
- **Architectural Shift:**
  - NiN avoids fully connected layers altogether.
  - Instead, NiN uses a NiN block with a number of output channels equal to the number of label classes.
  - Followed by a global average pooling layer, yielding a vector of logits.

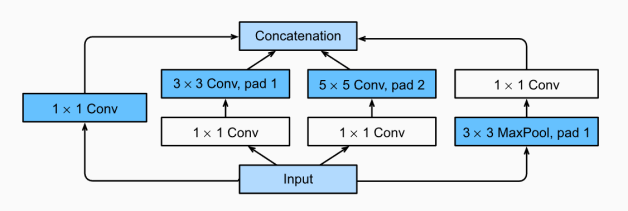


GoogLeNet: Multi-Branch Networks

Szegedy, C. et al. (2015). Going deeper with convolutions. Proceedings of the CVPR (pp. 1–9).

Network Structure:

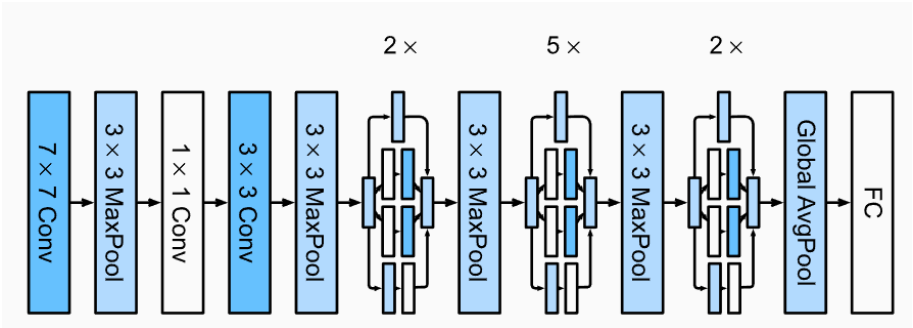
- **Stem for data ingestion:**
  - First two or three convolutions.
  - Operate directly on the image.
- **Body for data processing:**
  - Contains convolution blocks.
  - It has concatenated multi-branch convolutions.
- **Head for prediction:**
  - Maps features to the required problem.
  - Handles classification, segmentation, detection, or tracking.



Convolution block in GoogLeNet is the **Inception Block**.

GoogLeNet Model

GoogLeNet uses a stack of a total of 9 inception blocks, arranged into three groups with max-pooling in between, and global average pooling in its head to generate its estimates. Max-pooling between inception blocks reduces the dimensionality. At its stem, the first module is similar to AlexNet and LeNet.



Applying Batch Normalization in CNN

- **Placement:** With convolutional layers, we can apply batch normalization after the convolution but before the nonlinear activation function.
- **Per-Channel Logic:** The key difference from batch normalization in fully connected layers is that we apply the operation on a per-channel basis across all locations.
- **Dimensions:**
  - Assume minibatches contain m training samples.
  - For each channel, output height is p and width is q.
  - Normalization is carried out over the $m \cdot p \cdot q$ elements per output channel simultaneously.
- **Spatial Normalization:**
  - Collect values over all spatial locations to compute mean and variance.
  - Apply the same mean/variance within a given channel to every spatial location.
- **Parameters:** Each channel has its own scale and shift parameters, both of which are scalars.

Other Advanced CNN Architecture

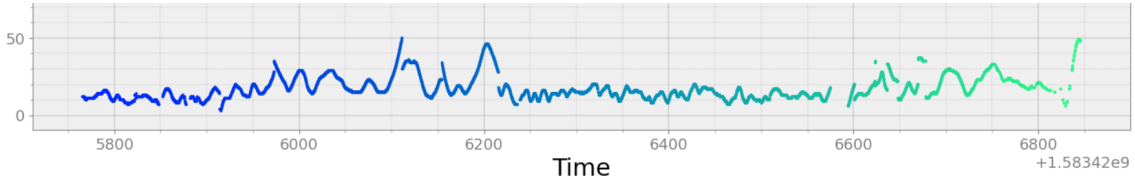
1. Xception - Deep Learning with Depthwise Separable Convolutions https://arxiv.org/abs/1610.02357
2. DenseNet - Densely Connected Convolutional Neural Networks https://arxiv.org/abs/1608.06993v5
3. MobileNetV1 - Efficient Convolutional Neural Networks for Mobile Vision Applications https://arxiv.org/abs/1704.04861v1
4. MobileNetV2 - Inverted Residuals and Linear Bottlenecks https://arxiv.org/abs/1801.04381
5. ConvNeXt - A ConvNet for the 2020s https://arxiv.org/abs/2201.03545

Recurrent Neural Network

Motivation

We have seen that Convolutional Neural Networks are effective in capturing spatial relationships in data like images. Further, CNN is good for classification, and regression like tasks are complicated to implement with CNN.

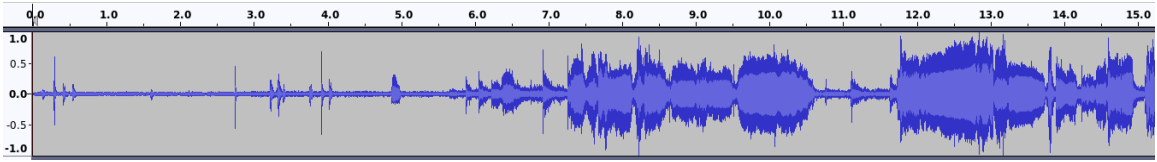
However, for data where sequential relationships are important (e.g., text, time series), we need something else.



Timeseries Data.

Some Example of Sequence Data

Speech Signal:



Text Sequences:

გამარჯობა! მე მიყვარს საქართველო - ჩვენი სილამაზით განთქმული ქვეყანა. საქართველო მდებარეობს კავკასიაში, შავი ზღვის აღმოსავლეთ სანაპიროზე. ჩვენი ქვეყანა ცნობილია

Language Model

Definition

A **language model** is a machine learning model designed to predict upcoming words in a sequence.

1. It assigns a **probability** to each possible next word, providing a *probability distribution*: $P(w_t \mid w_1, \ldots, w_{t-1})$.
2. It can also assign a joint probability to an **entire sentence**: $P(w_1, \ldots, w_n)$.

Which has a higher probability of appearing?

"all of a sudden I notice three guys standing on the sidewalk"

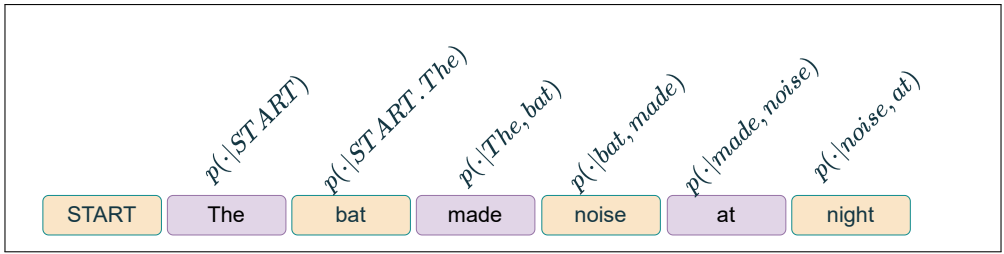
vs.

"on guys all I of notice sidewalk three a sudden standing the"

N-Gram Language Model

Core Objective

- Simplest kind of Language Model.
- **Goal:** Generate realistic-looking sentences in human language.
- **Key Idea:** Condition on the last $n-1$ words to sample the n-th word.



n-Gram Language Model

Question: How can we define a probability distribution over a sequence of length T?

The owl made noise at night

w_1 w_2 w_3 w_4 w_5 w_6

n - Gram Model (n = 2)

Using chain rule:

$$p(w_1, w_2, ..., w_T) = p(w_1)p(w_2|w_1)p(w_3|w_{1:2}) \cdots p(w_n|w_{1:n-1})$$

$$p(w_1, w_2, ..., w_T) = \prod_{t=1}^{T} p(w_t|w_{t-1})$$

$$P(w_1, w_2, w_3, w_4, w_5, w_6) =$$

- $p(w_1)$ (The)
- $p(w_2|w_1)$ (The $\rightarrow$ owl)
- $p(w_3|w_2)$ (owl $\rightarrow$ made)
- $p(w_4|w_3)$ (made $\rightarrow$ noise)
- $p(w_5|w_4)$ (noise $\rightarrow$ at)
- $p(w_6|w_5)$ (at $\rightarrow$ night)

n-Gram Language Model

Question: How can we define a probability distribution over a sequence of length T?

The owl made noise at night

w_1 w_2 w_3 w_4 w_5 w_6

n - Gram Model (n = 3)

$$p(w_1, w_2, ..., w_T) = \prod_{t=1}^{T} p(w_t | w_{t-1}, w_{t-2})$$

$$P(w_1, w_2, w_3, w_4, w_5, w_6) =$$

- $p(w_1)$ (The)
- $p(w_2|w_1)$ (The $\rightarrow$ owl)
- $p(w_3|w_2, w_1)$ (The, owl $\rightarrow$ made)
- $p(w_4|w_3, w_2)$ (owl, made $\rightarrow$ noise)
- $p(w_5|w_4, w_3)$ (made, noise $\rightarrow$ at)
- $p(w_6|w_5, w_4)$ (noise, at $\rightarrow$ night)

Learning an n-Gram Model

Question: How do we learn the probabilities for the n-Gram Model?
Answer: Calculate relative frequencies from training data:

$$p(w_t|w_{t-n+1},\ldots,w_{t-1}) = \frac{\text{count}(w_{t-n+1},\ldots,w_t)}{\text{count}(w_{t-n+1},\ldots,w_{t-1})}$$

3-Gram Counts (context: "cows eat"):

Next Word	Count
grass	2
hay	2
corn	3

Example: 3-Gram Probabilities
Training corpus with agricultural sentences:

- ... cows eat grass ... (appears 2 times)
- ... cows eat hay daily ...(appears 2 times)
- ... cows eat corn ... (appears 3 times)

Calculation: $p(\text{corn}|\text{cows eat}) = \frac{3}{7} \approx 0.43$

Probability Distribution:

Next Word	Probability
grass	0.29
hay	0.29
corn	0.43

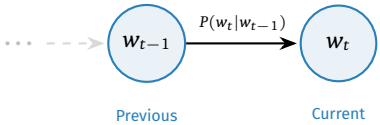
Markov Assumption

Definition

The assumption that the probability of a word depends only on the previous word is called a Markov assumption.

Markov Models

Markov models are the class of probabilistic models that assume we can predict the probability of some future unit without looking too far into the past.



Visualizing the first-order dependency.

How to estimate probabilities

Maximum Likelihood Estimation method:
Count words from a corpus, and normalize the counts so they fall between 0 and 1.
Computing 2-gram probability:

$$p(w_n|w_{n-1}) = \frac{C(w_{n-1}w_n)}{\sum_w C(w_{n-1}w)} = \frac{C(w_{n-1}w_n)}{C(w_{n-1})}$$

because the sum of all 2-gram counts that start with a given word w_{n-1} is equal to 1-gram count for that word w_{n-1}.

Example

```
<s> I am Sam </s>
<s> Sam I am </s>
<s>I do not  like green peas and cheese</s>
```

$P(I| < s >) = 2/3 = 0.67, P(< s > |Sam) = 1/2 = 0.5, P(Sam| < s >) = 1/3 = 0.33, P(Sam|am) = 1/2 = 0.5, P(am|I) = 2/3 = 0.67, = 0.5P(do|I) = 1/3 = 0.33$

Sampling from a Language Model

Process:

1. Start with initial context (n-1 words)
2. Roll weighted die for next word
3. Slide window and repeat

Sampling Demonstration:

Step	Context	Sampled Word
1	[START] The	bat
2	The bat	made
3	bat made	noise
4	made noise	at
5	noise at	night
6	at night	[END]

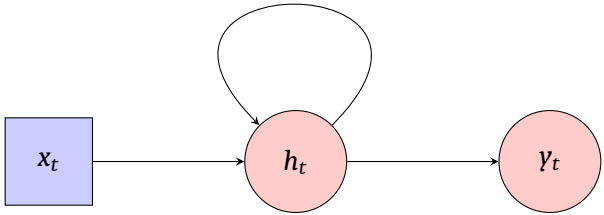
Generated Sentence:

The | bat | made | noise | at | night

Sampling introduces diversity but preserves local coherence through n-gram constraints.

Recurrent Connections and Recurrent Neural Networks

Recurrent Connection: Introduction of a cycle in a network.
Recurrent Neural Network (RNN): Any neural network that contains a cycle within its network connection.
There are many types of RNN. We will see the **Elman network** (1990, Elman).



Elman Network

The equations for the Elman network are:

$$h_t = g(Uh_{t-1} + Wx_t)$$
$$y_t = f(Vh_t) \tag{3}$$

where g is an activation function (e.g., tanh or sigmoid).
At the end, output layer also contains softmax: $y_t = \text{softmax}(Vh_t)$.

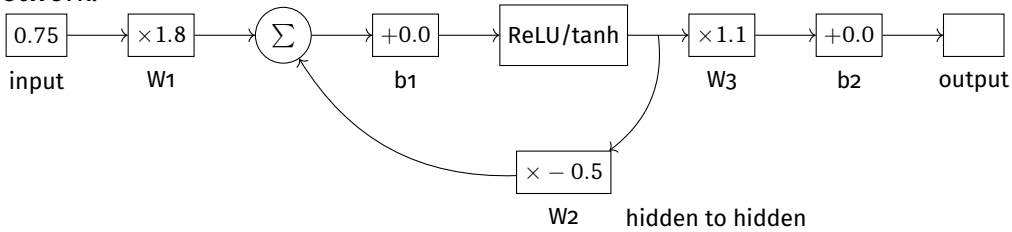
The goal is to learn the weights U, W, and V.

At the end output layer, softmax is often used.

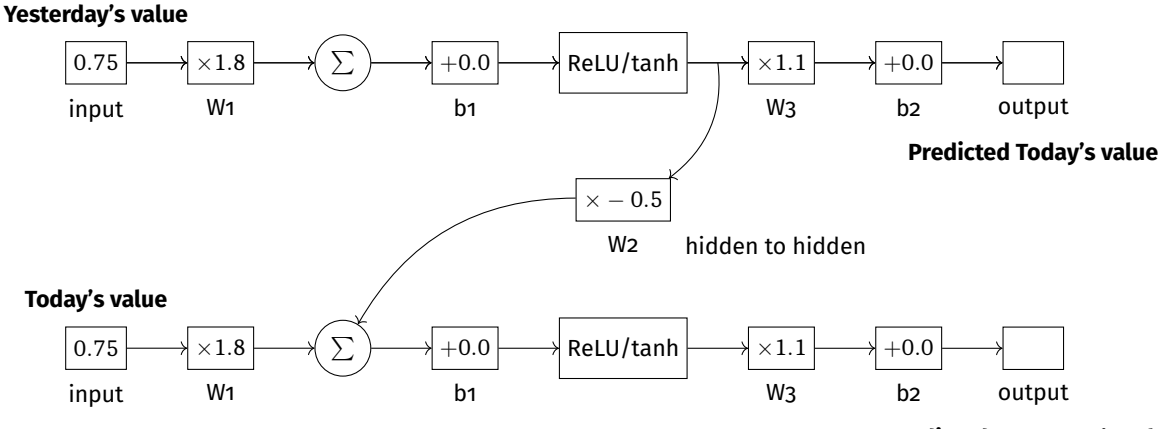
See an Example RNN Network

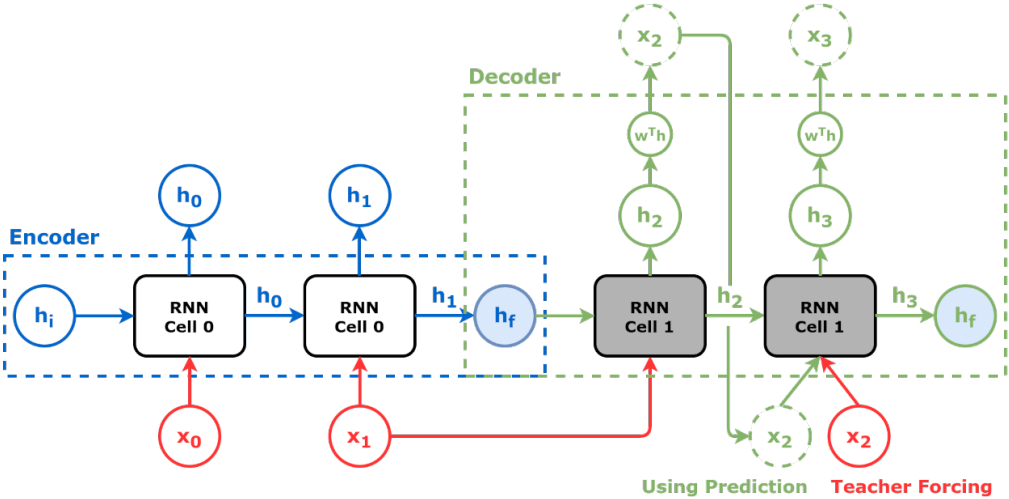
The feedback loop in Recurrent Neural Networks (RNNs) networks makes it possible to establish long-range dependence on sequential data.

To understand how these networks process sequential data over time, we can ”unroll” the network.



Unrolling the RNN Network





Source: https://github.com/dvgodoy/dl-visuals

Teacher Forcing

The idea that the predicted value is not used as input but the true value is used is called **teacher forcing**.

Vanishing Gradient / Exploding Gradient Problem

As we see that when we unroll the network, we have multiplication of W_2 (recurrent weight) with itself many times.

Exploding Gradient

If $W_2 > 1$, then eventually, the gradient will become really large ($W_2^{50} \gg 1$).

Vanishing Gradient

If $W_2 < 1$, then eventually, the gradient will become really small ($W_2^{50} \ll 1$).

Vanishing Gradient / Exploding Gradient Problem

Consider an RNN with ReLU activation:

$$h_t = \max(0, W_1 x_t + W_2 h_{t-1} + b_1)$$
$$y_t = W_3 h_t + b_2 \tag{4}$$

The backpropagation through time (BPTT) is used for training sequence data. The gradient of the loss (L) with respect to W_2 is:

$$\frac{\partial L}{\partial W_2} = \sum_{k=1}^{t} \frac{\partial L}{\partial y_k} \cdot \frac{\partial y_k}{\partial h_k} \cdot \frac{\partial h_k}{\partial W_2}$$

where k is the index for time.

Vanishing Gradient / Exploding Gradient Problem

$$\frac{\partial h_k}{\partial W_2} = \frac{\partial h_k}{\partial h_{k-1}} \cdot \frac{\partial h_{k-1}}{\partial W_2} = W_2 \cdot \frac{\partial h_{k-1}}{\partial W_2} = W_2 \cdot \left(W_2 \cdot \frac{\partial h_{k-2}}{\partial W_2} \right) = W_2^2 \cdot \frac{\partial h_{k-2}}{\partial W_2}$$

Continuing this pattern, we get:

$$\frac{\partial h_k}{\partial W_2} = W_2^{k-1} \cdot \frac{\partial h_1}{\partial W_2}$$

This shows that the gradient $\frac{\partial h_k}{\partial W_2}$ is proportional to W_2 and the previous gradients.

As you see, if $|W_2| > 1$, then the gradient grows exponentially (exploding gradient).

If $|W_2| < 1$, then the gradient vanishes.

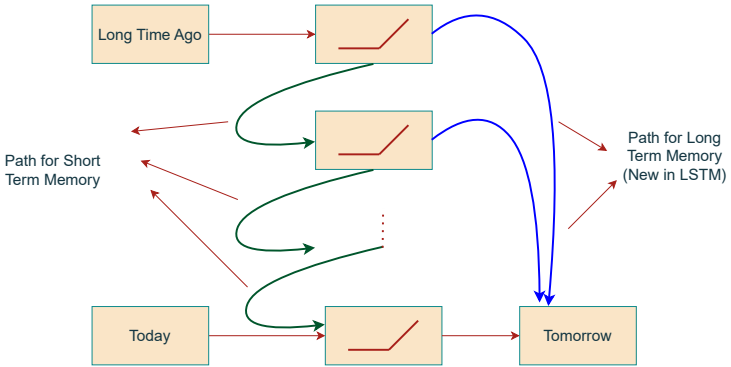
Long Short-Term Memory(LSTM)

In LSTM, we modify connections, so that there are two separate paths for long-term dependencies and short-term dependencies.

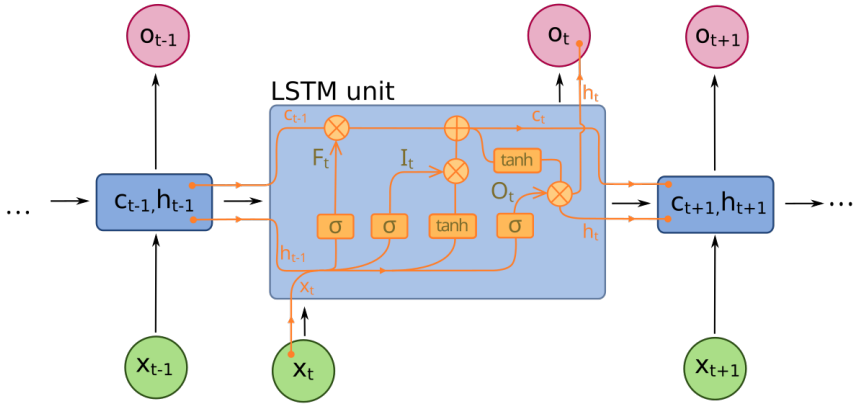
- LSTM specifically uses the tanh and sigmoid activation functions:

$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}}$$



LSTM



Source: https://github.com/dvgodoy/dl-visuals

What's special in LSTM

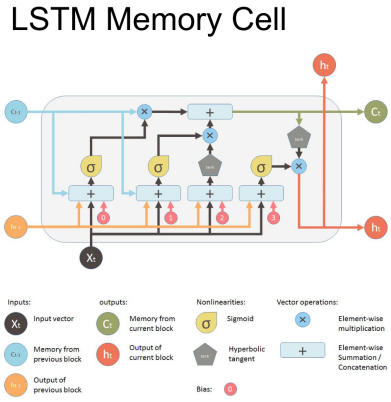
Central Idea

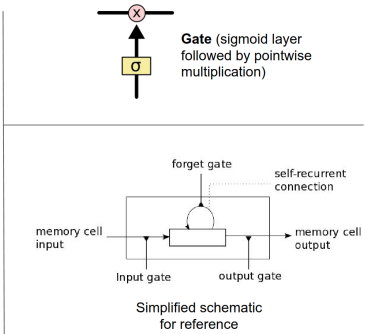
LSTM Introduces cell state and gating mechanisms to preserve long-term information. A memory cell (interchangeably block) which can maintain its state over time, consisting of an explicit memory (aka the cell state vector) and gating units which regulate the information flow into and out of the memory.

There are 3 kinds of gating mechanism:

- **Forget Gate**: Decides what information to discard
- **Input Gate**: Decides what new information to store
- **Output Gate**: Decides what to output

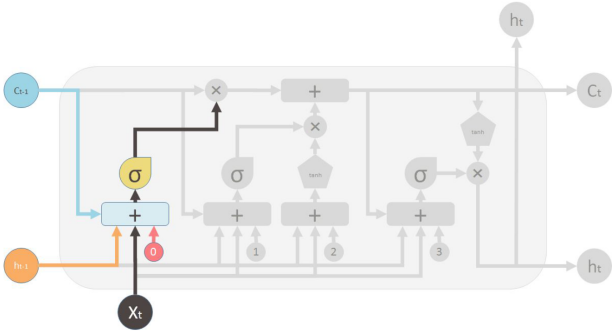
LSTM Memory Cell



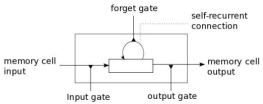


Source: https://pages.cs.wisc.edu/~shavlik/cs638/lectureNotes/Long%20Short-Term%20Memory%20Networks.pdf

Forget Gate



$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] \; + \; b_f\right)$$



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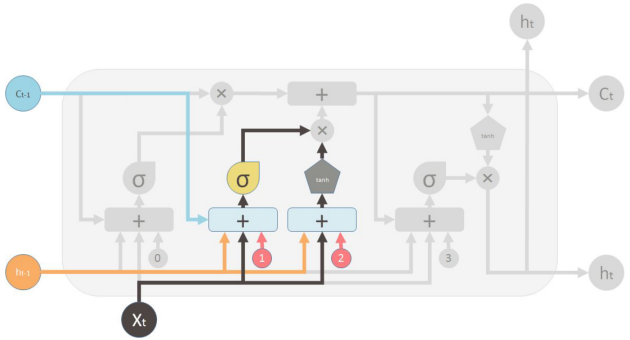
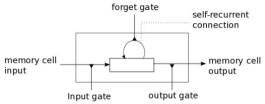
Input Gate



$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] \; + \; b_i\right)$$

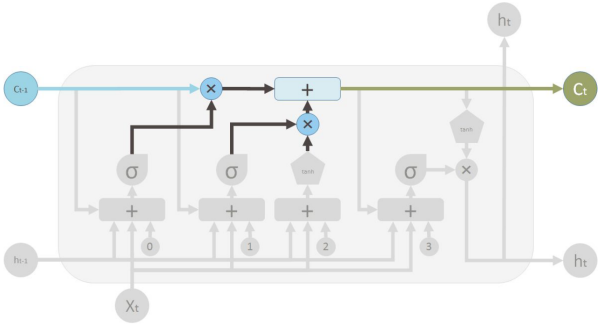
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] \; + \; b_C)$$



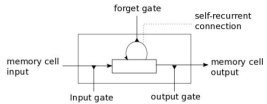
Source: https://pages.cs.wisc.edu/~shavlik/cs638/lectureNotes/Long%20Short-Term%20Memory%20Networks.pdf

Memory Update

The cell state vector aggregates the two components (old memory via the forget gate and new memory via the input gate)

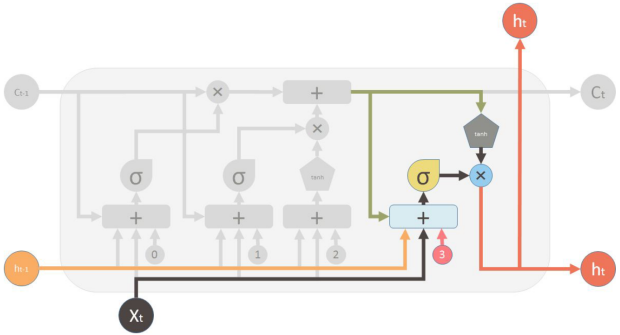


$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$



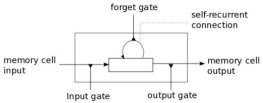
Source: https://pages.cs.wisc.edu/~shavlik/cs638/lectureNotes/Long%20Short-Term%20Memory%20Networks.pdf

Output Gate



$$o_t = \sigma\left(W_o\ [h_{t-1}, x_t]\ +\ b_o\right)$$

$$h_t = o_t * \tanh\left(C_t\right)$$



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References

1. https://web.stanford.edu/~jurafsky/slp3/3.pdf